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Stability of Magnetization in Cainozoic Basalts from Victoria, Australia

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Abstract

Instability of magnetization in basalts is shown to be dependent on the composition of the magnetic mineral, which is a titanomagnetite. Low Curie temperature titanomagnetites tend to acquire secondary components of magnetization while high Curie temperature titanomagnetites of composition near to magnetite do not.

§ 1. Introduction

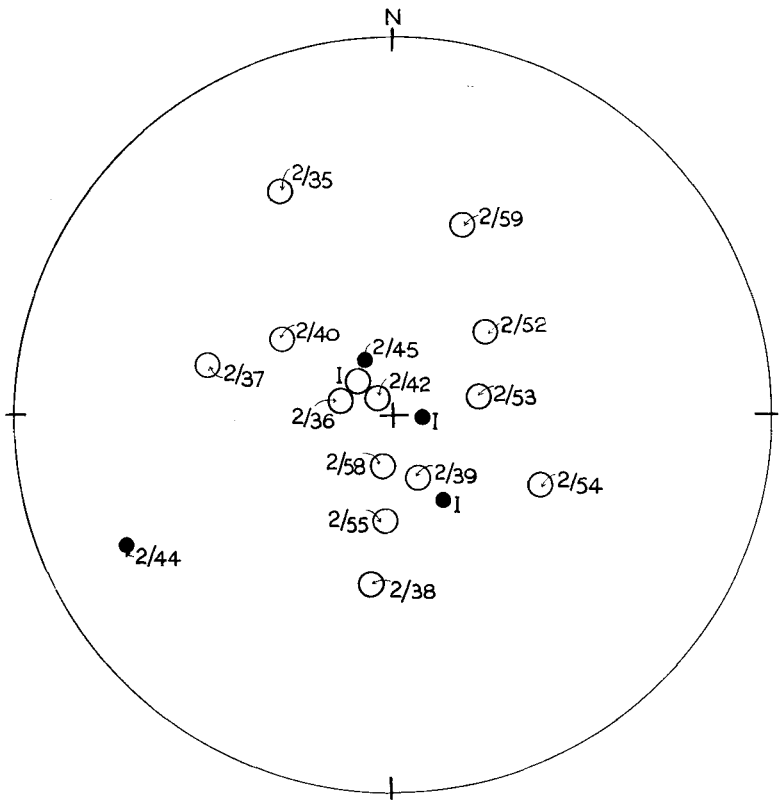
Localities of mixed polarities of magnetization were reported by Green and Irving (1958) in their investigation of the palaeomagnetism of the Cainozoic volcanics of Victoria, Australia. At Irving's suggestion the author re-sampled one of these sites Drouin South, in considerable detail and examined the directions of magnetization of the rocks more fully in order to see if a more complete picture of the mixed polarity could be obtained.

Cylinders of rock were prepared from the samples collected, in the usual way, and the directions of magnetization were measured using an astatic magnetometer. The directions of magnetization that were obtained by measurements made at the University of Adelaide are shown in fig. 1. This figure also shows the directions of magnetization determined by Irving and Green. It can readily be seen from this figure that the directions obtained by the author do not fall into two groups 180° apart as was originally thought to be the case, but are much more scattered, indicating the possibility of instability in the magnetization.

Lavas, on cooling through their Curie point, acquire a primary thermoremanent magnetization (TRM) in the direction of the earth's field. If magnetic particles with relaxation times less than the geological age of the rock are present, a secondary component, an isothermal remanent magnetization (IRM) is acquired differing in direction from the primary one if the direction of the geomagnetic field has changed since the TRM was acquired.

This of course represents an unwanted unstable component of magnetization which must be removed if the rocks are to yield any useful information.

Fig. 1



Stereographic projection showing original magnetization directions at Drouin South.

○ Point on upper hemisphere.
● Point on lower hemisphere.

Those points marked I are directions determined by Irving and Green.

§ 2. A. C. Demagnetization Experiments

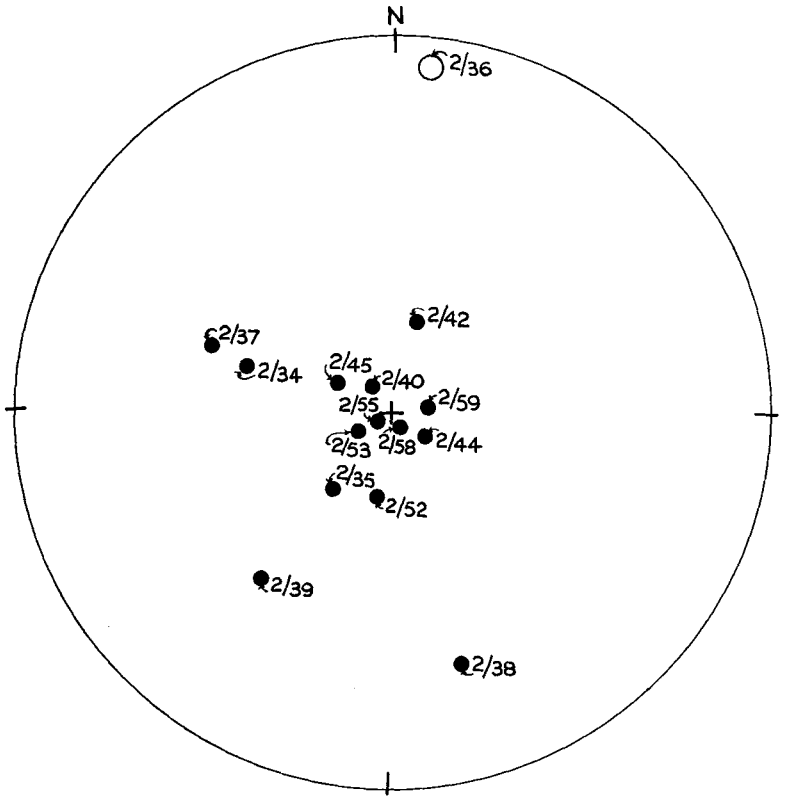
The author carried out a.c. demagnetization experiments on the specimens taken from Drouin South at the Department of Geophysics, A.N.U., Canberra, where a.c. washing apparatus similar to that described by Creer (1958) was made available to him.

By subjecting rock specimens to gradually decreasing alternating magnetic fields it is possible to remove unwanted, unstable components of magnetization which may have been acquired since the formation of the rocks. The a.c. washing in this case was done in peak fields of 150 and 225 oersted which were gradually reduced to zero.

After washing, the directions of magnetization of all except one of the thirteen specimens which were originally normally magnetized became

reversed. The initially reversed specimens remained reversed. These directions of magnetization are plotted in figs. 2 and 3, and the values of the precision factor K (Fisher 1953) for these two distributions are given in table 1.

Fig. 2



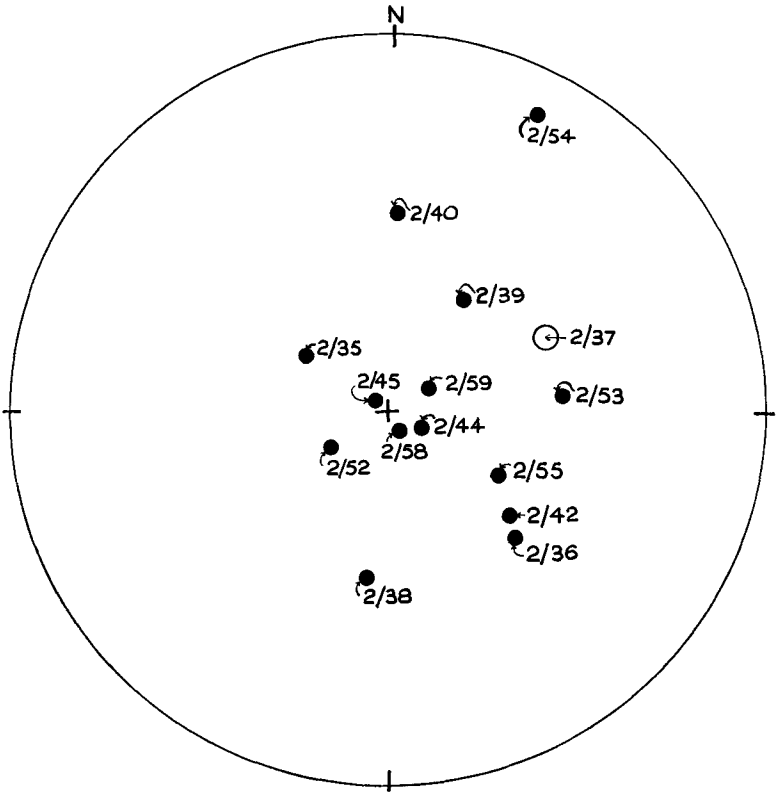
Stereographic projection showing directions of magnetization after washing in 150 oersted.

○ Point on upper hemisphere.
● Point on lower hemisphere.

Table 1

Washing field (oersted)	K	Mean direction of magnetization	
		D	I
150	4·4	256	+80
225	3·2	75	+71

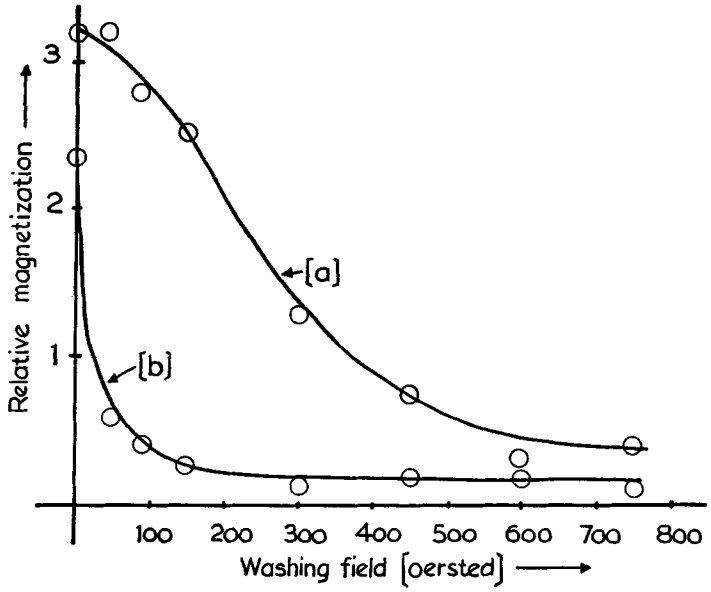
Fig. 3



Stereographic projection showing directions of magnetization after washing in 225 oersted.

○ Point on upper hemisphere.
● Point on lower hemisphere.

Fig. 4



Two of the specimens, one originally reversed (2/45) and one normally magnetized (2/59), were washed progressively in fields of 45, 90, 150, 300, 450, 600, 750 oersted. The a.c. demagnetization curves for these two specimens are shown in fig. 4. From these two curves it is seen that the magnetization of the reversed specimen is the more stable.

These a.c. demagnetization experiments show that the mixed polarities which do exist at Drouin South are the result of a temporary unstable component of magnetization superimposed on a stable reversed one.

§ 3. Thermal Demagnetization Experiments

In order to see if there was any difference in the behaviour of these specimens to thermal demagnetization, as well as to the a.c. demagnetization already described, specimens from the rocks were heated in an inert atmosphere of nitrogen, in zero magnetic field inside a non-magnetic furnace constructed by the author. The variations of the horizontal components of magnetization of the rocks with temperture were measured using an astatic magnetometer mounted above the furnace. This investigation was carried out at the University of Adelaide. Typical thermal demagnetization curves are shown in fig. 5. These curves indicate that there are both single and multi-phase magnetic systems present in these rocks. From the thermal demagnetization curves it was concluded that the magnetization of these basalts is due to a titanomagnetite, the composition of which varies considerably even in this one quarry, causing the rocks to have very different thermal demagnetization properties.

The Curie points for the specimens examined are given in table 2.

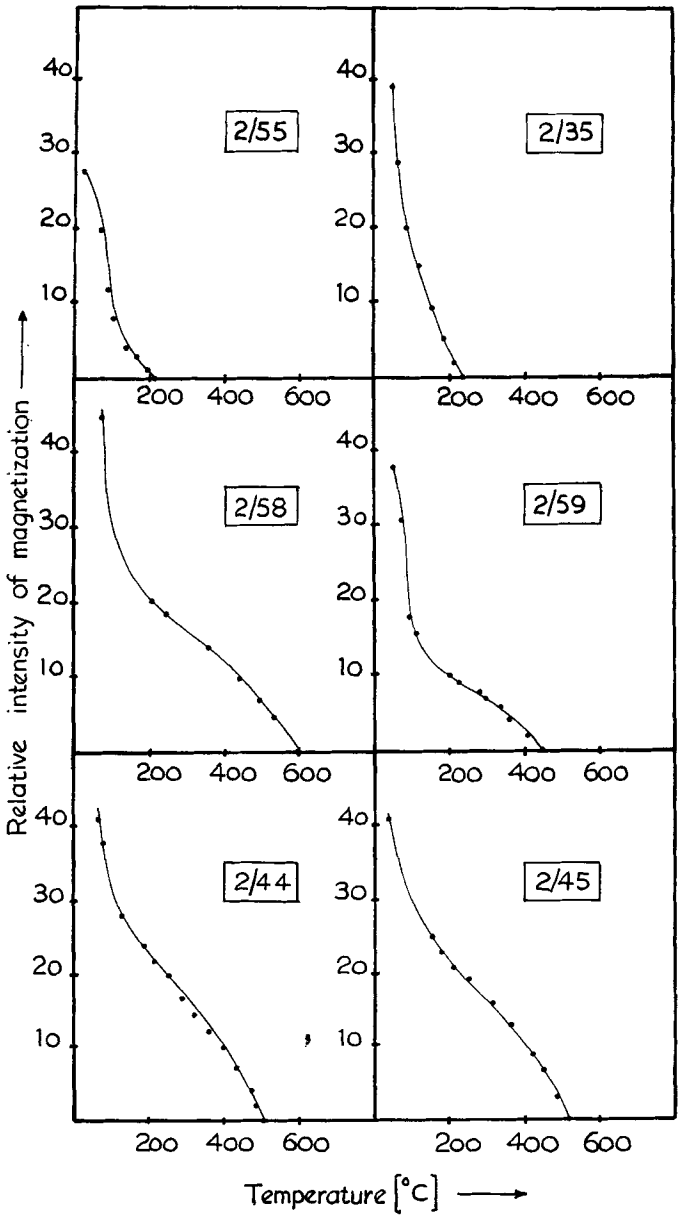
Table 2

Specimen	Curie point (°c)
2/52	300
2/54	260
2/55	220
2/56	330
2/35	230
2/37	310
2/40	260
2/41	430
2/58	300, 580
2/59	175, 435
2/44	225, 500
2/45	400, 510

Further specimens from some of those samples for which thermal demagnetization curves were obtained were also washed progressively in peak fields of 45, 90, 150 oersted using apparatus similar to that described by

Irving *et al.* (1961) which was constructed by the author at the University of Adelaide. The change in the directions of magnetization after washing in these fields is given in table 3 together with the Curie point obtained.

Fig. 5



Typical thermal demagnetization curves for specimens collected at Drouin South. Numbers 2/44 and 2/45 are the two specimens which retained their originally reversed direction.

It is seen that the stability of magnetization is dependent on the Curie temperature of the main magnetic component present in the rock. That is, specimens 2/44, 2/45 (see fig. 5) contain a larger amount of high Curie point titanomagnetite than low Curie point titanomagnetite, and are the only two specimens which still retain their original direction of magnetization. The other rocks, including 2/58, 2/59 which contain mainly a low Curie point titanomagnetite, have acquired large unstable components of magnetization which can however be removed by a.c. washing as has already been described. The rocks containing a high Curie point titanomagnetite of the composition near magnetite possess a stable hard magnetization, those which contain a low Curie point titanomagnetite possess an unstable soft magnetization.

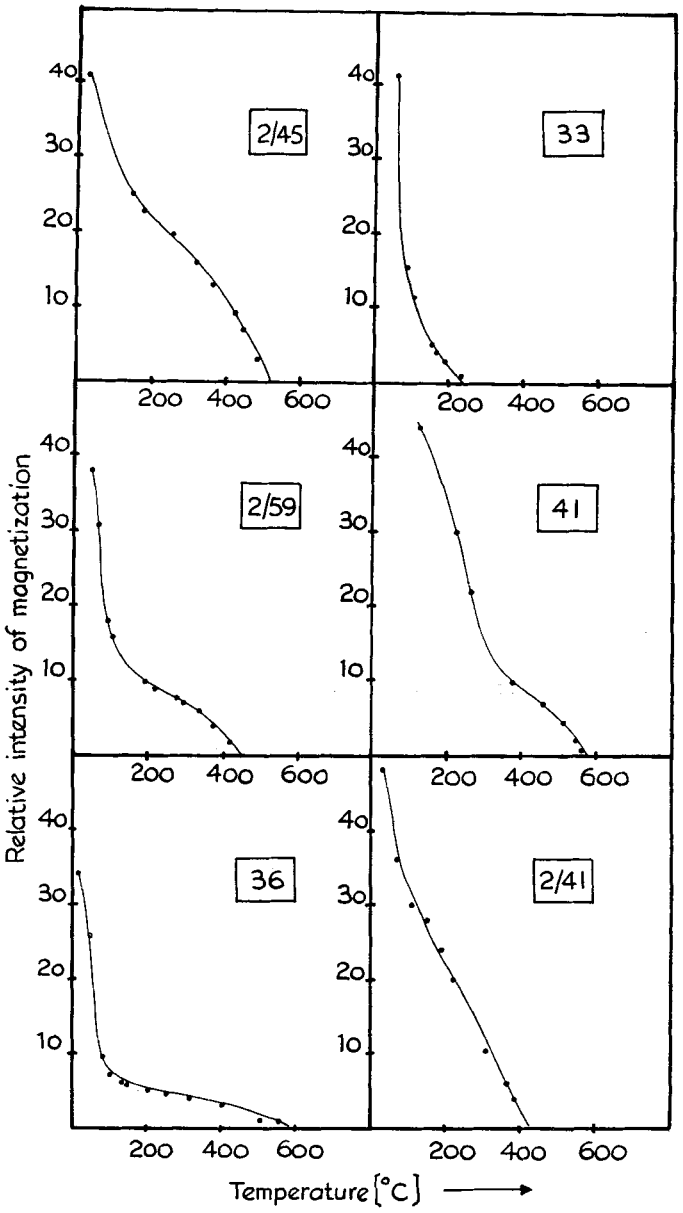
Table 3

Specimen	Curie point (°c)	0		45		90		150	
		D	I	D	I	D	I	D	I
2/41a	430	265	−8	217	+75	143	+80	60	+44
2/44a	225, 500	233	+30	204	+76	222	+84	76	+80
2/52a	300	34	−55	50	−56	85	−20	292	+18
2/54a	260	106	−38	0	−62	27	−11	− 5	+54
2/55a	220	195	−72	207	−55	44	+56	−77	+55
2/56a	330	148	−83	92	−75	68	−38	68	+26
2/58a	300, 580	191	−74	181	+66	122	+80	123	+80
2/59a	175, 435	21	−34	18	+33	25	+69	47	+74
2/45a	400, 510	338	+66	6	+72	314	+80	324	+80

The results of further thermal and a.c. demagnetization experiments which show that stability is dependent on the composition of the titanomagnetite in the rocks are shown in figs. 6, 7, 8. In fig. 6 there are the thermal demagnetization curves for specimens 2/45, 2/59, 2/41 taken from Drouin South, together with those for specimens 33, 36 and 41 which are also specimens collected from the Older Volcanic basalts of Victoria.

In fig. 7 there are given the a.c. demagnetization curves for specimens from these rocks, while in fig. 8 there are curves which show $\Delta M/\Delta H$ for ΔH equal to intervals of 50 oersted. ΔM is the change of relative magnetization in the interval ΔH. $\Delta M/\Delta H$ therefore itself is only a relative measure of the rate of change of magnetization with washing field, and need only be, for the important point to be determined from these curves is the a.c. washing field at which the rate of change of magnetization is greatest. It is seen from these curves that the stability of the magnetization to a.c. demagnetization seems to depend on the Curie temperature of the main titanomagnetite which the basalt contains, as the rate of change of magnetization is greater at small washing fields for those specimens which contain the low Curie point titanomagnetites.

Fig. 6

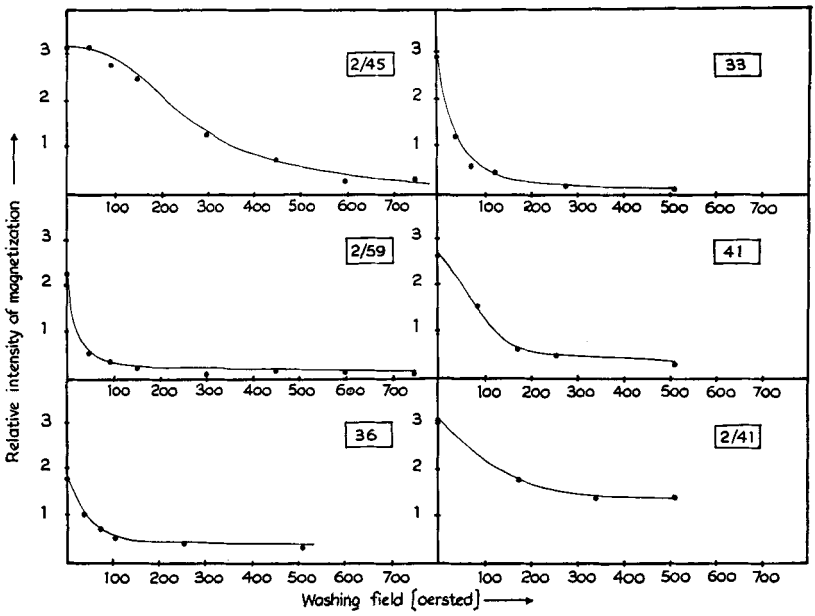


Thermal demagnetization curves for specimens of basalt referred to in text.

§ 4. A Discussion of Stability of Magnetization in Basalts

There has been increasing literature in recent years on the stability of magnetization in rocks and perhaps the most up-to-date investigation of this type carried out on igneous rocks has been reported on by Irving *et al.* (1961). They have given a satisfactory method for treating igneous rocks to remove unstable components of magnetization, and this is the method used in § 2 . They also discuss how the rocks retain the stable NRM and suggest that the stable magnetization is retained in high stability regions within the ferromagnetic grains, and that the explanation for the hard and soft magnetic components which are found not only in basalts, but also in other igneous rocks is to be sought in exsolved phases of differing grain size in ferromagnetic grains.

Fig. 7



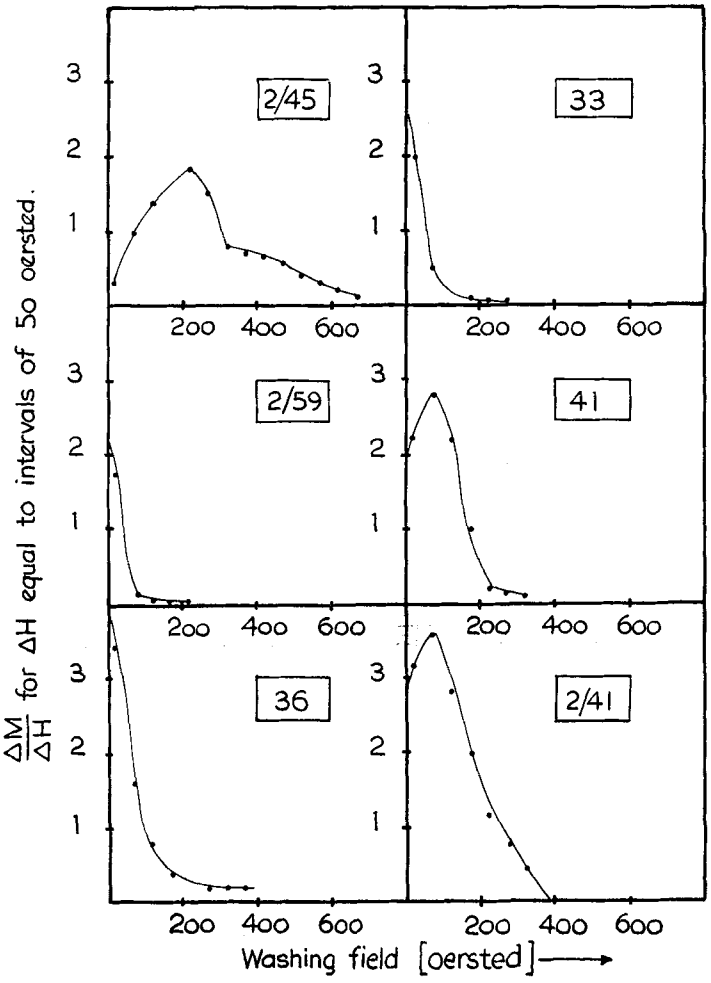
A.C. demagnetization curves for specimens whose thermal demagnetization curves are given in fig. 6.

The magnetic minerals in most basalts are titanomagnetites, as is the case with these specimens collected from Drouin South. A relation between stability and Curie temperature of the titanomagnetites in these basalts has been indicated by the a.c. and thermal demagnetization experiments described in §§ 2 and 3, and such a relation explains in part the findings of the previous investigations by Irving *et al.* (1961).

The procedure outlined by Irving *et al.* (1961) to remove unstable components of magnetization involves washing the rock specimens in

gradually increasing peak fields. and choosing the distribution of directions of magnetization which has the least scatter. They also report that after treatment in high alternating fields, the intensity of magnetization of a specimen does not diminish regularly, with increasing alternating field,

Fig. 8

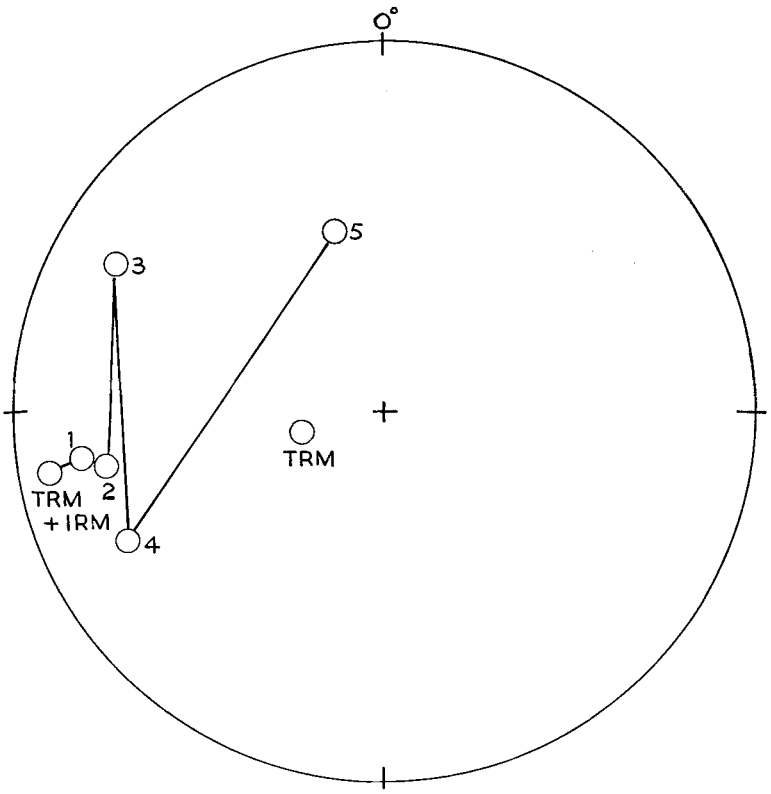


Curves showing relative values of $\Delta M/\Delta H$ (ΔH equal to 50 oersted intervals), for the a.c. demagnetization curves shown in fig. 7.

but levels off to almost a constant average value when directions of magnetization become nearly random. This value is called the minimum intensity ($M_{\min}$) of magnetization. They suggest that the cause for this is that the spontaneous magnetization of domains is lost only at temperatures above the Curie point, that the effect of alternating magnetic fields is one

of rearrangement of domains, and because of statistical fluctuations the resultant moment of a specimen is never likely to be zero.

Fig. 9



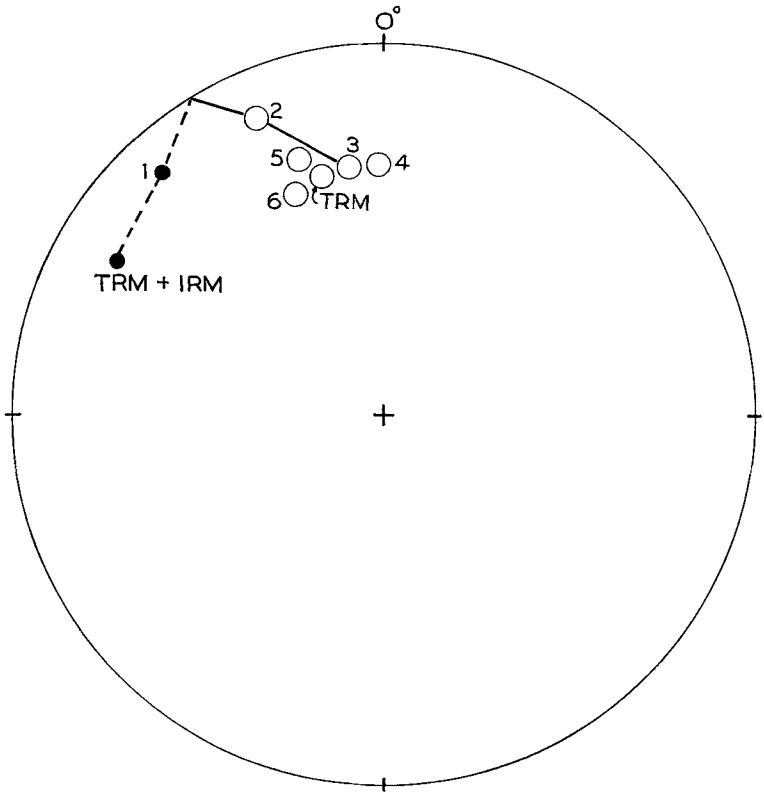
Stereographic projection showing the change in direction of magnetization in specimens 2/55 when it is washed in increasing a.c. fields. The specimen has been given a TRM in 0·6 oersted together with an IRM in 100 oersted.

(1) 45, (2) 90, (3) 170, (4) 260, (5) 340 oersted.

The reason why different washing fields produce minimum scatter of direction in different sites is in large part due to the fact that the magnetic minerals present in one site may have Curie temperatures hundreds of degrees different from those present in another site, and so possess a different degree of instability. One of the reasons for the poor grouping of directions of magnetization after washing the specimens from a particular site is due to the fact that the specimens may have a large range of Curie temperatures, as they do at Drouin South, and so the stability varies widely within this

one site. While a specimen which has ferromagnetic grains composed mainly of magnetite probably does not acquire large temporary components of magnetization, quite large washing fields can be used to remove any temporary components of magnetization before the TRM is affected too much by the washing. On the other hand, those specimens which are composed mainly of a titanomagnetite with a low Curie temperature are

Fig. 10



Stereographic projection showing the change in direction of magnetization in specimen 2/44 when it is washed in increasing fields. The specimen has been given a TRM in 0·6 oersted together with an IRM of 100 oersted. (1) 45, (2) 90, (3) 170, (4) 260, (5) 340, (6) 430 oersted.

very likely to acquire large unstable components of magnetization, but only small washing fields can be used to demagnetize such specimens, for the magnetic moment reaches a minimum intensity of magnetization after washing in fairly small fields, and the directions of magnetization begin to become scattered, almost random. This type of behaviour is shown in figs. 9 and 10.

These two figures show the effect of washing two specimens, one which contains a very low Curie temperature titanomagnetite (2/55), the other which contains a high Curie temperature titanomagnetite (2/44), under the following circumstances. Both of the specimens have been given a TRM by heating them above their Curie temperatures, to 250°C and 500°C respectively, and allowing them to cool in the earth's field. They were then given an IRM in about 100 oersted and were washed in fields of 45, 90, 170, 260, 340 oersted. It is seen that in the case of the specimen which contains the high Curie temperature titanomagnetite, the IRM is removed by the washing, and that the direction of magnetization of the specimen returns to the direction of the original TRM. In the case of the rock containing the low Curie temperature titanomagnetite, however, after initial washing the direction of magnetization begins to return to the direction of magnetization of the TRM given to the specimen, but before the IRM is completely removed the TRM of the specimen itself begins to be affected by the increased washing field, and the direction of magnetization after washing in these higher fields becomes random.

Table 4

Specimen number	Curie temperature (°C)
2/55	225
2/58	300, 580
2/44	225, 500

It would seem then that not all these basalts contain a hard and soft component of magnetization, but that they, in many cases, possess only a magnetization which varies in its degree of hardness or softness. As far as those basalts which do possess both hard and soft magnetic components (that is high and low Curie temperature titanomagnetites) are concerned, Irving *et al.* (1961) have suggested that the accuracy of the procedure of removal of unwanted secondary magnetizations by alternating magnetic fields will be limited by the presence of the minimum intensity in those cases where the magnitude of the soft component is large, since the minimum intensity of this may exceed the magnitude of the hard component.

To show what effect this minimum intensity of magnetization of a soft component can have in those cases where there are high Curie temperature, and low Curie temperature components present, the author has taken three specimens, the thermal demagnetization curves of which are given in fig. 5. They are specimens 2/55, 2/58 and 2/44. It is seen that two of these are two component types, one containing mostly a low Curie temperature titanomagnetite, the other containing mostly a high Curie temperature titanomagnetite. The third contains a low Curie point titanomagnetite only. The Curie temperatures of the various components in these specimens are given in table 4.

These specimens were washed repeatedly in a field of 500 oersted, and the direction and the moment of magnetization were measured after the washing. The washing was repeated ten times for each specimen in this same field of 500 oersted. In a field of this magnitude, specimen 2/55 and the low Curie temperature titanomagnetite of 2/58 and 2/44 should have reached the minimum intensity condition, but the high Curie temperature material of specimens 2/58 and 2/44 should not yet have reached this condition, as is indicated by fig. 4, the a.c. demagnetization curve for specimen 2/54 which contains a high Curie temperature titanomagnetite having a Curie point of 510°C compared with 500°C for specimen 2/44.

The direction of magnetizations and the moment of the specimen after each washing is given in table 5.

Table 5

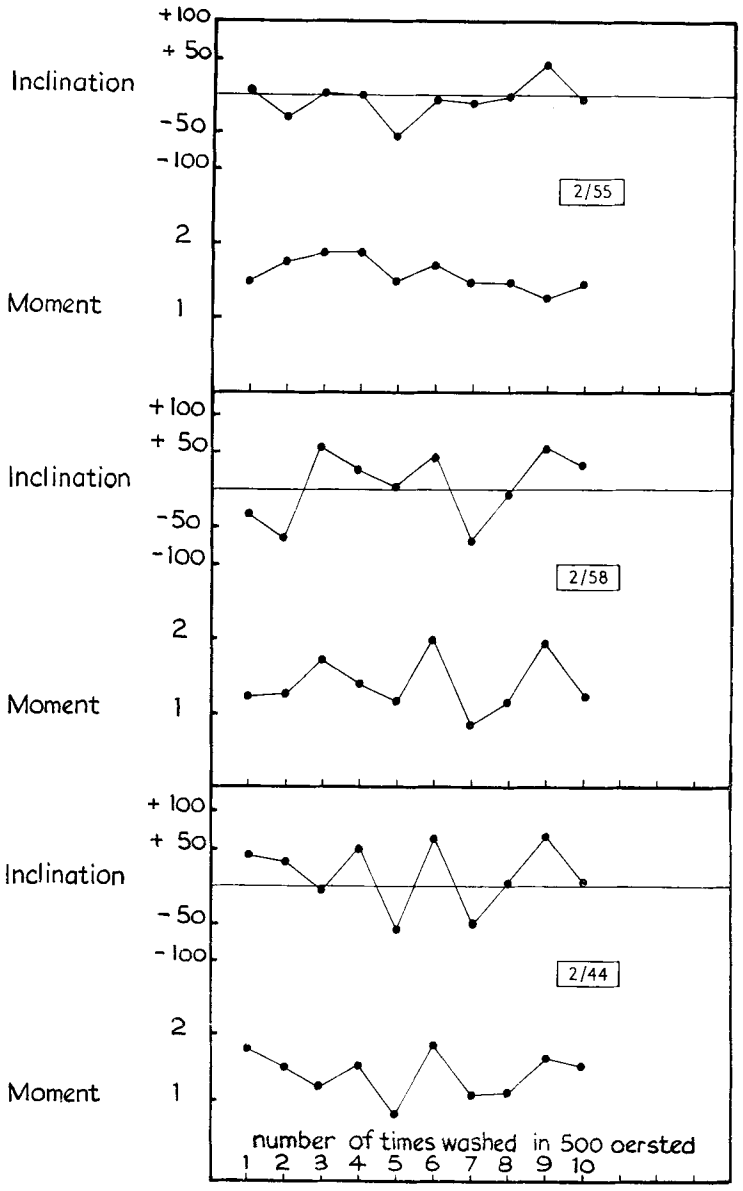
Specimen 2/55b			Specimen 2/58b			Specimen 2/44		
M	D	I	*M*	D	I	*M*	D	I
$1{\cdot}50\times10^{-2}$	217	+ 4	$1{\cdot}25\times10^{-2}$	161	−30	$1{\cdot}79\times10^{-2}$	187	+44
$1{\cdot}79\times$,,	194	−30	$1{\cdot}28\times$,,	133	−61	$1{\cdot}55\times$,,	− 3	+34
$1{\cdot}90\times$,,	−112	+ 5	$1{\cdot}74\times$,,	−86	+54	$1{\cdot}28\times$,,	42	− 2
$1{\cdot}91\times$,,	− 75	0	$1{\cdot}42\times$,,	− 6	+24	$1{\cdot}59\times10^{-2}$	41	+46
$1{\cdot}48\times$,,	152	−56	$1{\cdot}18\times$,,	−10	+ 4	$0{\cdot}89\times10^{-3}$	−51	−58
$1{\cdot}72\times$,,	− 39	− 2	$2{\cdot}04\times10^{-2}$	−55	+46	$1{\cdot}86\times10^{-2}$	36	+61
$1{\cdot}48\times$,,	85	− 8	$8{\cdot}53\times10^{-3}$	−23	−67	$1{\cdot}17\times$,,	−43	−51
$1{\cdot}50\times$,,	− 34	− 1	$1{\cdot}15\times10^{-2}$	168	− 5	$1{\cdot}21\times$,,	161	+ 6
$1{\cdot}30\times$,,	−121	+45	$1{\cdot}95\times$,,	−86	+56	$1{\cdot}69\times$,,	126	+67
$1{\cdot}48\times$,,	86	− 3	$1{\cdot}26\times$,,	204	+36	$1{\cdot}59\times$,,	173	+ 4

In fig. 11 the angle of inclination and the magnetic moment are plotted against the number of the washing. It is quite readily seen that in the case of the specimens which contain both hard and soft components of magnetization, the magnetic moment varies as the direction of magnetization. When the direction of magnetization is reversed, the stable direction for the site determined in § 2, the magnetic moment is greatest. These results of course suggest that there is a reversed stable direction in the specimens which is not affected much by the repeated washing, together with a direction which varies with washing. In other words, there is a hard magnetization, the direction of which does not change greatly on washing together with a soft magnetization, the direction of which does vary considerably with washing because it has reached the minimum intensity condition in demagnetization, and the direction of magnetization takes near random positions after repeated washing in the same high field.

It is also evident from the results of the a.c. demagnetization experiments carried out on the rocks from Drouin South that some of the originally stable TRM is retained even in the very unstable low Curie point titano-

magnetites. It is possible that the reason why any of the original magnetization is retained may be due to the fact that there are many grains of ferromagnetic material present in these basalts which are $< 20\,\mu$ in size.

Fig. 11



The magnetic moment and the inclination of magnetization plotted against the number of times the specimen has been washed in an a.c. field of 500 oersted. For explanation see text.

These smaller grains may tend to retain the direction of the original TRM because of the large increase in the coercive force with decrease in grain size (Gottschalk 1935). Thus, while the lower Curie point titanomagnetites are less stable than the higher Curie point types with a composition near to that of magnitite, in these low Curie point types the direction of the original TRM is probably retained to some extent in the ferromagnetic particles of smaller grain size.

§ 5. Conclusions

From the investigation of the basalts taken from Drouin South it is clear that the stability of the magnetization of these rocks depends on the hardness of the magnetization which they possess. Those rocks which contain mainly magnetite are stable, those which contain mainly a low Curie temperature titanomagnetite are unstable. In the two component types which contain some magnetite, or near magnetite along with the low Curie temperature titanomagnetite, the stable magnetization is retained in the hard magnetic component. That is the higher Curie temperature component which has the composition near to magnetite itself, but this stable direction may be masked by the minimum intensity of the magnetization of the soft component after washing.

The instability in the lower Curie temperature titanomagnetite, or soft magnetic component may be removed by a.c. demagnetization as has been described by Irving *et al.* (1961). Some of the soft magnetic material, probably the larger grains present, easily acquires an isothermal remanent magnetization in the direction of the earth's field, which can be removed by the washing. What must be appreciated however is that while small washing fields can remove such a magnetization from these soft magnetic components, (the low Curie temperature titanomagnetites) quite low a.c. fields will reduce the TRM to its minimum intensity value when directions of magnetization become nearly random.

Acknowledgments

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